Practice Quiz 10A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Strong induction on a recursively defined sequence.

1. Let $a_1 = 2$, $a_2 = 5$, and $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 3$. Prove by strong induction that $a_n = 3n - 1$ for every integer $n \geq 1$. State the strong inductive hypothesis in full and check both initial cases.

   Solution We use strong induction. Let $n \geq 1$ and suppose

   $$a_k = 3k - 1 \quad \text{for every integer } k \text{ with } 1 \leq k < n.$$

   If $n = 1$, then $a_1 = 2 = 3(1) - 1$. If $n = 2$, then $a_2 = 5 = 3(2) - 1$.

   If $n \geq 3$, both $n - 1$ and $n - 2$ are at least 1 and less than n. The hypothesis therefore applies to both terms, so

   $$\begin{aligned} a_n &= 2a_{n-1} - a_{n-2} \\ &= 2(3(n-1) - 1) - (3(n-2) - 1) \\ &= 2(3n - 4) - (3n - 7) = 3n - 1. \end{aligned}$$

   All cases are covered. Strong induction proves the formula for every integer $n \geq 1$. ∎

Solutions revised using GPT-6 Astra in the Codex harness.